



Open Source as Institutional Infrastructure

Lessons from the UC OSPO Network

Tim Dennis, Director, Data Science Center
UCLA Library, Teaching, Learning & Research Department

Leigh Phan, Data Scientist, Data Science Center
UCLA OSPO Co-Leads, UC OSPO Network

The invisible layer

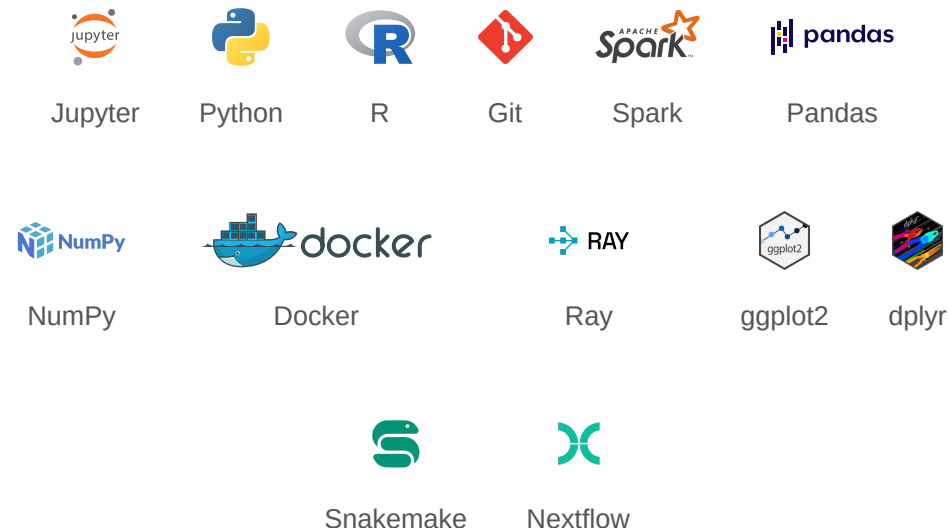
Research data workflows run on open source:

- Data collection, cleaning, analysis, visualization
- Reproducibility tooling: containers, workflows, version control
- Repositories and scholarly infrastructure

92% of researchers use software; **67%** say their research would be impossible without it.

The people keeping it running are increasingly called **Research Software Engineers (RSEs)**, a growing professional community with no standard institutional home.

If every organization had to replace open source with proprietary equivalents, the cost would exceed **\$8.8 trillion**.



We've been here before

People in this room helped build the open data movement:

- FAIR principles: from aspiration to institutional expectation
- Data management plans embedded in grant workflows
- Curation, provenance, and stewardship as professional services
- Open science infrastructure: repositories, PIDs, metadata standards
- FAIR4RS: FAIR principles now extended to research software

Software is the next frontier, and UC has been showing the way for 50 years: BSD, Unix, Ceph, Apache Spark, RISC-V, Ray. The pipeline from university research to global infrastructure is not rare or accidental.

“The research done in academia was the seed. Foundations provided the soil. But the community was the sun.”

Summer 2000. UC Santa Cruz.

PhD student Jim Kent assembled the human genome on 50 borrowed PCs, icing his wrists to keep coding.

He finished **three days** before a corporate competitor's supercomputer.

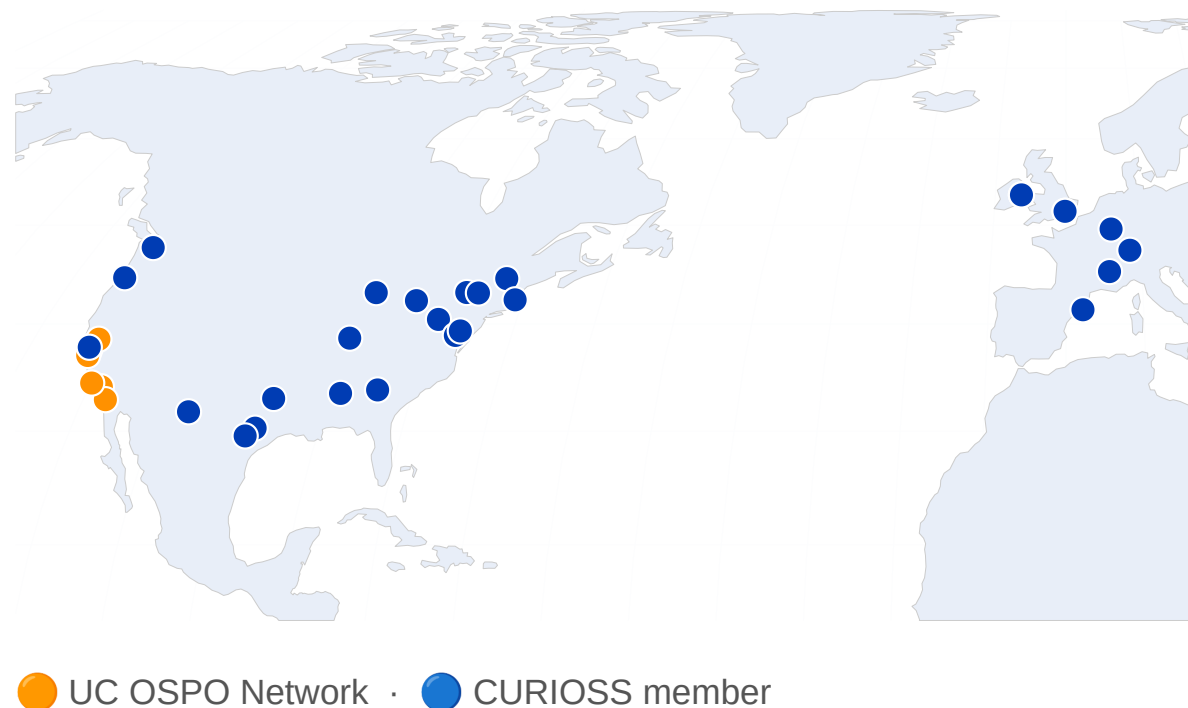
The genome stayed in the public domain.

Kent, J.W. (2002). BLAT - The BLAST-Like Alignment Tool. *Genome Research* 12, 656–664. · genome.ucsc.edu

What is an academic OSPO?

An **Open Source Program Office** supports, governs, and promotes open source within an institution.

- Originated in industry: ~77% of large tech companies have one
- First academic OSPOs: Johns Hopkins, UC Santa Cruz, and RIT (2021)
- **34 academic OSPOs** in CURIOSS worldwide and growing
- Core functions: policy, licensing guidance, best practices, education, community



The UC OSPO Network

A **multi-campus collaboration** treating open source as shared infrastructure.

- 6 campuses: Santa Cruz, Berkeley, Davis, Los Angeles, Santa Barbara, San Diego
- Launched April 2024, funded by the Alfred P. Sloan Foundation
- **\$1.85M** in external funding to date
- Lead: UC Santa Cruz, the first OSPO in a large state university system
- Serves 280,000+ students and 25,000+ faculty

Acts as a **neutral convener**: no single campus, company, or grant cycle can pull the work away.



Three thematic areas



Discovery

Mapping who does what
across the UC system:
repos, contributors, tools,
and practices



Sustainability

Keeping projects and
communities healthy
through licensing, project
health, and community
support



Education

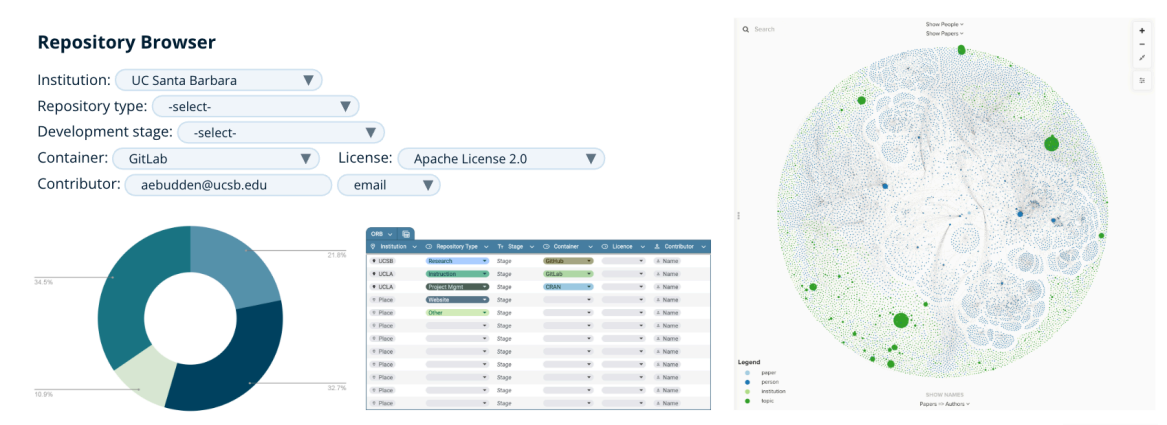
Coordinated training and
learning pathways across
campuses

Discovery: knowing what's there

The 4 Ps framework, applied at system scale:

- **People:** who contributes to OS software at UC?
- **Product:** what has been built?
- **Practice:** how do faculty and staff engage?
- **Perception:** what are contributors' experiences?

Tools: GitHub pipeline (200,000+ repos scanned; ~52,000 institutionally affiliated across 10 campuses), network-wide survey (294 respondents), UC Open Repository Browser (UC ORB)



UC Open Repository Browser showing filter controls, repository type breakdown, data table of UC campus repos, and a network visualization of project relationships

Sustainability: keeping it running

58% of UC open source contributors are also maintainers. Not users: stewards.

What they asked for most: sustainability grants, computing infrastructure, learning communities.

#1 challenge: **finding time to write documentation.**

Nationally: **60%** of maintainers are unpaid; **43%** report burnout.

Network services:

- Cross-campus licensing working group (UCOP + tech transfer + IT)
- Project health assessment (OSSPREY)
- Containerization support · Community management

58%

of UC OSS contributors
are also **maintainers**
not just users, stewards

60%

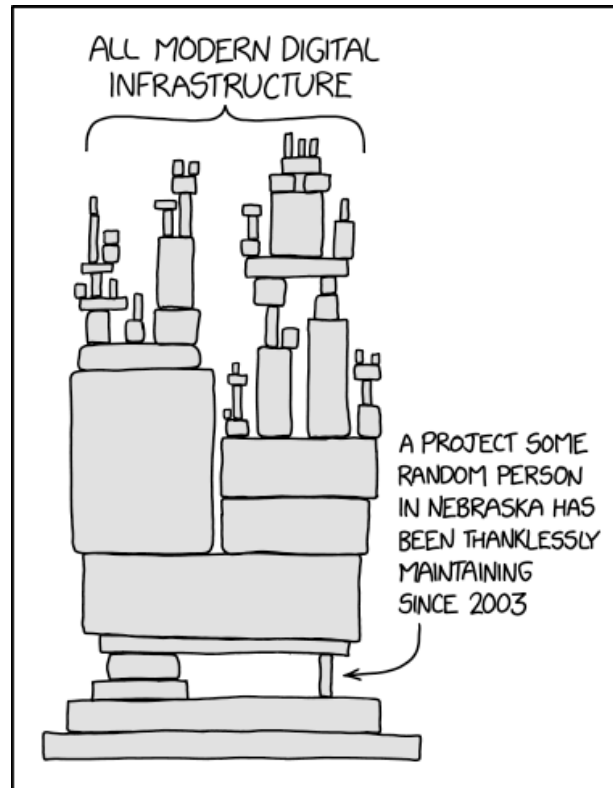
unpaid nationally

43%

report burnout

Tidelift (2024) State of the Open Source Maintainer Report

The dependency problem



XKCD #2347 'Dependency': a towering stack of blocks labeled 'All modern digital infrastructure' balances on a single tiny block labeled 'A project some random person in Nebraska has been thanklessly maintaining since 2003'. By Randall Munroe.

xkcd #2347 (CC BY-NC 2.5) · UC survey: 58% of contributors are maintainers · Tidelifit (2024): 60% unpaid, 43% burnt out

AI and open source: a new pressure

AI code generation is changing what it means to maintain a project.

- **Slop PRs:** AI-generated contributions that look complete but are off-spec, shallow, or carry subtle bugs
- Review burden grows while maintainer capacity stays flat
- The OSS projects that trained the models are now flooded by their output
- Security risk: AI-generated code in dependencies with undetected vulnerabilities

OSPOs are developing responses: contribution disclosure policies, AI-aware code review norms, project-level guidance.

Baltes, Cheong & Treude (2026) arxiv:2603.27249 · GitClear (2024) “Coding on Copilot”

The structural shift

70% of new AI PhDs go straight to industry

(a decade ago: 50/50)

90% of notable AI models come from a handful of companies

Academia is being sidelined from the systems it helped create.

Ruff (2026) UC Open Summit ·
youtu.be/eBriL3CDNeo

Education: training as infrastructure

Gap analysis → curriculum inventory → learning pathways

- Inventoried existing materials: Carpentries, CodeRefinery, The Turing Way, UC-specific content
- Organized by topic (licensing, sustainability, community) and by role (contributor, maintainer, manager)
- Published at ucospo.net/education
- Coordinated with data literacy programs and Carpentries infrastructure

UC OSPO NETWORK

- About
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 - All Pathways
 - All Lessons
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 - Search
 - Develop a Lesson
 - For Educators

Welcome to the Open Source Learning Pathways.

What are you here to learn today?

Getting Started with Open Source
For learners new to open source and open collaboration.

Open Source Literacy

Beginner
GUIDE
CONTRIBUTOR
Finding and Evaluating Projects
Learn how to find your first project using discovery tools, understanding labels, and identifying beginner-friendly tasks.
Updated 12d ago

Beginner
WORKSHOP
COMMUNITY MANAGER, MAINTAINER
Social Coding and Inclusive Communication
Understand the 'human' side of open source, including respectful language, the 'release early, often' mindset, and the value of non-cod...
Updated 3mo ago

Beginner
WORKSHOP
IP, GOVERNANCE
Software Licensing
This lesson provides an overview of common open-source software licenses, their conditions, and how to choose the right one for your project.
Updated 7d ago

UC OSPO Network Education site showing the Open Source Learning Pathways, organized by role, skill level, and pathway, with beginner lesson cards including Software Licensing, Social Coding, and Finding and Evaluating Projects

Distributed expertise, system-wide capacity

Each campus brings what others don't. The network routes it to where it's needed.

Each campus contributes

- Unique domain expertise and staff knowledge
- Local relationships with researchers and faculty
- Pilots that can only run with campus context
- Specialized capacity others can learn from

The network multiplies it

- Expertise flows across all six campuses
- Shared staffing no single campus could fund
- Smaller campuses build capacity they couldn't alone
- A success at one becomes a template for five

Governed by an OSPO Leadership Group and thematic working groups, one per campus, coordinated at system level.

Our seat is scholarly communication

We already promote best practices across the research lifecycle. Software is the same work in a new medium.

What we do for scholarship	The software version
Citation and attribution	software citation, CITATION.cff
Persistent identifiers	Zenodo DOIs for releases
Metadata and discovery	repository metadata, FAIR4RS
Licensing and rights	open source license guidance
Preservation	versioned, archived releases

Not a new mission. The same one, pointed at code.

Where this meets your work

For data professionals, the OSPO sits at the intersection of:

- Research software engineering ↔ data management
- Licensing compliance ↔ open data policy
- Reproducibility tooling ↔ FAIR / FAIR4RS
- Training infrastructure ↔ data literacy programs
- RSE community ↔ library research support

For your institution:

- Where does open source software governance currently live?
- What would it take to treat it as infrastructure rather than a side project?

You're already doing this work

If you do technical data services, researchers are already bringing you these questions:

- “What license should I use for this code?”
- “Is this library still maintained? Should I depend on it?”
- “How do I make this reproducible and citable?”
- “My funder wants a software management plan.”
- “How do I accept contributions from collaborators?”

That is OSPO work. The frame makes it legible to you, your institution, and the researchers you serve.

Raise the floor, not just solve the ticket

Fixing their immediate problem gets them through the week.

Moving them from “it runs on my machine” to maintainable, licensed, citable software gets their whole lab there, and their students after them.

You don't need a formal OSPO to do this. You need the frame.

OSPO heuristics already in your toolkit: health signals · bus factor · governance files · FAIR4RS · software DMPs · contributor conventions · community governance

Where it fits: instruction and outreach

Two tracks, both already in our wheelhouse.

Informational (the why)

- Why cite software, how to choose a license, planning for sustainability
- Workshops, consultations, guides, outreach

Technical practice (the how)

- README, LICENSE, CITATION.cff, a versioned release, a DOI
- Hands-on: the UC OSPO citable-software lesson

We don't have to build an OSPO to take a seat. We teach the practices.

What you can do right now

If you advise on data-intensive research, you are probably already helping with code.

The network has already built the resources (Laura Langdon, UC OSPO):

- [README.md](#) - template + guide
- [LICENSE](#) - guide includes UC-approved options
- [CONTRIBUTING.md](#) - template + guide
- [CODE_OF_CONDUCT.md](#) - Contributor Covenant + guidance
- [CITATION.cff](#) + Zenodo release → citable, versioned DOI

Going further:

- [Growing a Community Around a Project](#) - UC OSPO
- [Sharing Research Software: Reproducible, Citable, and Discoverable](#) - Carpentries Incubator lesson (covers [pixi](#), [CITATION.cff](#), Zenodo, discoverability)

Learn more / connect

UC OSPO Network: ucospo.net

- Education site: ucospo.net/education
- Mailing list, Slack, events calendar



UC OSPO Network logo

Global network: curiOSS.org, 34 academic OSPOs worldwide and growing

ucospo.net

Tim Dennis Director, Data Science Center, UCLA Library tdennis@library.ucla.edu

Slides: [link TBD]

Barker, M., Chue Hong, N. P., Katz, D. S., Lamprecht, A.-L., Martinez-Ortiz, C., Psomopoulos, F., Harrow, J., Castro, L. J., Gruenpeter, M., Martinez, P. A., & Honeyman, T. (2022). Introducing the FAIR principles for research software. *Scientific Data*, 9, 622.

<https://doi.org/10.1038/s41597-022-01710-x>

Brown, C. T., & Irber, L. (2016). Sourmash: A library for MinHash sketching of DNA. *Journal of Open Source Software*, 1(5), 27.

<https://doi.org/10.21105/joss.00027>

Cerf, V. G., & Kahn, R. E. (1974). A protocol for packet network intercommunication. *IEEE Transactions on Communications*, 22(5), 637–648. <https://doi.org/10.1109/TCOM.1974.1092259>

Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1), 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>

Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., Wang, L.-P., Simonett, A. C., Harrigan, M. P., Stern, C. D., Wiewiora, R. P., Brooks, B. R., & Pande, V. S. (2017). OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7), e1005659. <https://doi.org/10.1371/journal.pcbi.1005659>

Gomez, J., Lovell, E., Lieggi, S., Cardenas, A. A., & Davis, J. (2025). *Recipe for discovery: A pipeline for institutional open source activity*. <https://arxiv.org/abs/2506.18359>

Hattrick, S., Antonietti, M., Carr, L., Chue Hong, N., Crouch, S., De Paiva, D., Emley, J., Gable, C., Hay, A., Inupakutika, D., et al.